Density-One Descent in the Collatz Map

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Abstract

We study the Collatz map via the fully accelerated transformation

$$f(x) = \frac{3x+1}{2^{v_2(3x+1)}},$$

where $v_2(n)$ denotes the 2-adic valuation of n.

Trajectories of f are encoded by finite division words (sequences of 2-adic division counts), each of which induces an affine transformation acting uniformly on entire congruence classes (cylinders). This representation yields a combinatorial framework in which admissible division words are generated by a constrained symbolic grammar that restricts local valuation growth patterns.

We establish bounds on the growth of admissible words, showing that their combinatorial expansion is strictly subcritical relative to the dyadic scaling imposed by 2-adic valuations. As a consequence, the set of integers that do not admit a finite descent prefix has natural density zero, with exponentially decaying tail estimates.

It follows that the set of integers that eventually attain a strictly smaller value under iteration of f has natural density one. This set admits a canonical disjoint decomposition into residue classes associated with minimal descent words, providing a measure-theoretic description of typical Collatz behavior.

While zero-density exceptional trajectories are not excluded, the result gives a rigorous structural description of global descent in the Collatz dynamics.

Contents

1 Introduction

The $3x + 1$ problem, also known as the Collatz conjecture [1], concerns the behavior of the map

$$T(x) = \begin{cases} \dfrac{x}{2} & \text{if } x \text{ is even,} \\ 3x + 1, & \text{if } x \text{ is odd.} \end{cases}$$

on the positive integers. It asserts that every positive integer eventually reaches 1 under iteration of T. Despite its elementary formulation, the conjecture has resisted proof for decades [2, 3]. Extensive computational verification confirms convergence for very large ranges [4], but does not illuminate the global structure of trajectories.

A major advance by Tao [5] showed that almost all integers, in the sense of logarithmic density, eventually decrease under iteration of the Collatz map. This result is obtained via probabilistic averaging rather than an explicit combinatorial decomposition of the integers.

The present work develops a deterministic alternative based on a symbolic and measure-theoretic decomposition of the integer lattice. Rather than analyzing individual trajectories, we encode dynamics using division words under the fully accelerated map. These words induce natural residue classes (cylinders), which provide a convenient framework for organizing density calculations.

We introduce two equivalent dynamical formulations of the Collatz map. The *semi-accelerated map*

$$g(x) = \begin{cases} \dfrac{x}{2} & \text{if } x \text{ is even,} \\ \dfrac{3x+1}{2} & \text{if } x \text{ is odd.} \end{cases}$$

and the *fully accelerated map*

$$f(x) = \frac{3x + 1}{2^{v_2(3x+1)}},$$

where $v_2(n)$ denotes the 2-adic valuation. The map f compresses each odd step of g into a single affine transformation and acts as the natural symbolic encoding space for trajectory structure.

Trajectories under f are encoded by finite *division words*, recording the 2-adic division counts after each odd step. Each division word induces an affine map describing the action of f on an entire arithmetic progression simultaneously. These words define residue classes modulo powers of two, which we identify canonically with dynamical *cylinders*.

This identification allows us to organize Collatz dynamics through nested families of division words and their associated cylinders. Minimal stopping-time conditions select a disjoint subfamily of cylinders corresponding to first descent events, whose total measure governs the global distribution of convergent behavior.

To control the combinatorial proliferation of admissible division words, we introduce an increment grammar encoding minimal 2-adic growth constraints. This grammar governs the structure of admissible cylinders and allows quantitative comparison between combinatorial growth and exponential measure decay, showing that combinatorial growth is strictly dominated by the exponential scaling imposed by valuations.

Using this framework, we show that the density contributions of convergent cylinders are summable with vanishing tails, so that the complement has natural density zero. Consequently, the union of convergent cylinders has natural density one. Thus almost every positive integer eventually attains a strictly smaller value under iteration of the Collatz map in the sense of natural density.

While zero-density exceptional trajectories are not excluded, the results provide a structural and measure-theoretic description of typical Collatz dynamics via a canonical cylinder decomposition of the integers.

Structure of the Paper. The proof proceeds through the following steps:

- Section 2 introduces stopping time and residue class construction.
- Section 3 develops division words and their affine encodings.
- Section 4 establishes the increment grammar governing admissibility.
- Section 5 constructs the horizontal-sum enumeration of cylinders.
- Section 6 establishes the finite-state linear structure of the horizontal-sum dynamics.
- Section 7 proves that the density of non-convergent integers vanishes.
- Section 8 completes the argument by establishing global descent in natural density.

2 Definitions and Preliminaries

2.1 Stopping Time

Definition 2.1. The *stopping time* $\sigma(x)$ of a positive integer x (sometimes called the finite stopping time or local descent time) is the minimal integer $k \geq 1$ (if such a k exists) such that

$$f^k(x) < x.$$

If no such k exists, we write $\sigma(x) = \infty$. Stopping time behavior has been studied extensively in the context of the $3x + 1$ problem [3].

2.2 Convergent Segments

Definition 2.2. A *convergent segment of length* k is a finite sequence

$$x, f(x), f^2(x), \ldots, f^k(x)$$

such that $f^k(x) < x$ and k is minimal. No assumption is made about eventual convergence to 1; only strict descent at step k, i.e., $k = \sigma(x)$, is required. Finite segments of this type were considered in the context of the accelerated Collatz map by Everett [6].

2.3 Residue Classes

Encoding trajectories via residue classes or arithmetic progressions has appeared in prior structural approaches to the $3x + 1$ problem [7, 8].

Iterates are indexed starting from 0, so that $f^0(x) = x$. Integers x and y are equivalent modulo depth k if the sequence of parities of $f^i(x)$ and $f^i(y)$ agree for $i = 0, \ldots, k-1$. In other words, they share the same parity sequence.

Definition 2.3 (Residue Class of Depth k)**.** Fix $k \in \mathbb{N}$ and $x \in \mathbb{N}$. Let $\mathrm{par}(n) = n \bmod 2$. The *residue class of depth k with respect to f* containing x is

$$[x]_k := \{y \in \mathbb{N} \mid \mathrm{par}(f^i(y)) = \mathrm{par}(f^i(x)) \text{ for } i = 0, 1, \ldots, k-1\}.$$

Here $[x]_k$ encodes the parity sequence; the associated *division word* ω further records the number of divisions by two following each iterate of f, as formalized in Section 3. Parity sequences provide a coarse combinatorial classification, while division words provide a refined encoding of the full 2-adic structure of the trajectory.

Remark (Arithmetic structure of residue classes)**.** Later (Section 4), we show that division words refine these parity-defined residue classes into arithmetic progressions modulo powers of two determined by the total division count $D(\omega)$.

At this stage, it suffices to know that $[x]_k$ encodes a parity sequence; any further arithmetic structure will be derived in subsequent sections.

Definition 2.4 (Convergent Residue Class)**.** A residue class $[x]_k$ is called *convergent of stopping time* k if the stopping time is constant over the class, i.e.

1. $f^k(y) < y$ for all $y \in [x]_k$, and
2. no smaller integer $j < k$ satisfies $f^j(y) < y$ for any $y \in [x]_k$.

Here k is minimal with this property for elements in $[x]_k$.

Remark. The uniformity of the inequalities across the class $[x]_k$ is not immediate from the definition of residue classes. It will follow from the affine representation of division words developed in Section 3, and will be formalized in Section 4.

Remark (Refinement by division words)**.** The residue classes $[x]_k$ defined above depend only on parity data and thus provide a coarse partition of $\mathbb{N}$. In later sections, division words ω will refine these classes by incorporating full 2-adic valuation data.

This refinement yields arithmetic progressions of the form

$$x \equiv r_\omega \pmod{2^{D(\omega)}},$$

which we will represent using the notation C_ω (cylinders) for convenience in density calculations. Thus the notation r_ω and C_ω represents a finer, valuation-based description of the same underlying structure.

3 Division Counts and Path Encoding

The structure of accelerated Collatz trajectories is governed by the total number of divisions by two. Residue classes arise as a convenient way to organize the resulting arithmetic constraints.

At each step of the accelerated map

$$f(x) = \frac{3x+1}{2^{v_2(3x+1)}},$$

all powers of two are removed from $3x+1$. The number of removed factors of two at the i-th iterate is therefore the 2-adic valuation

$$d_i := v_2(3f^i(x)+1).$$

Here d_i corresponds to the division count in the transition $f^i(x) \to f^{i+1}(x)$. Since the fully accelerated map always produces odd iterates, $3f^i(x)+1$ is even and hence $d_i \geq 1$.

3.1 Formal Definition of Division Words

In this section we introduce a symbolic encoding of Collatz trajectories based on the number of divisions by two removed at each step of the accelerated map. We call such an encoding a *division word*. The precise combinatorial constraints determining which division words correspond to actual trajectories are deferred to Section 4; however, the definitions given here are sufficient for the analysis of residue class densities.

Let a Collatz path be a sequence of iterates under the accelerated map

$$f(x) = \frac{3x+1}{2^{v_2(3x+1)}}.$$

Definition 3.1 (Division Word)**.** A *division word* of length $m \geq 0$ is a finite sequence

$$\omega = (d_0, d_1, \ldots, d_{m-1}),$$

where each d_j is a positive integer representing the number of divisions by two applied at the j-th iterate of the accelerated map. Admissibility constraints on which sequences can occur are defined in Section 4.

For $m = 0$, the word is empty and represents the *trivial zero-step segment*.

Lemma 1 (Consistency of Division Counts for Realizing Trajectories)**.** Let $\omega = (d_0, \ldots, d_{m-1})$ be a division word realized by an integer x. Then the sequence of valuations along the trajectory of x agrees with ω, i.e., $d_i = v_2(3f^i(x) + 1)$ for all $0 \le i < m$.

Remark. Uniformity of division counts across an entire residue class will follow from the affine representation of division words developed in this section, and will be formalized in Section 4.

3.2 Affine Maps Associated to Division Words

Each division word formally determines an affine map of any integer x:

$$x \longmapsto \frac{3^m x + c(\omega)}{2^{d_0+\cdots+d_{m-1}}}, \tag{3.1}$$

obtained by symbolic composition of the accelerated map along the word. Here $c(\omega)$ is a canonical integer uniquely determined by the division word (see Appendix A). This representation encodes the arithmetic effect of the sequence of accelerated steps encoded by the word on any starting integer.

Remark (Division Words Encode Structure, Not Values)**.** Fixing a division word ω determines the sequence of arithmetic operations applied along an accelerated Collatz trajectory. Consequently, both the linear coefficient $3^m/2^{d_0+\cdots+d_{m-1}}$ and the affine term $c(\omega)$ depend only on ω and not on the starting integer.

This allows reasoning about convergence or divergence at the level of residue classes without reference to specific integers.

Remark (Residue Classes and Division Words)**.** Each admissible division word ω induces a unique arithmetic progression modulo $2^{D(\omega)}$, provided the associated integrality condition is solvable. Admissibility conditions ensuring non-emptiness are developed in Section 4.

3.3 Total Division Count

Let a division word $\omega = (d_0, \ldots, d_{m-1})$ encode a segment consisting of m iterates of the accelerated map f. Each iterate of the fully accelerated map removes a finite number of factors of two from $3x + 1$. This number is precisely the entry of the division word corresponding to that iterate.

Definition 3.2 (Total Division Count)**.** For a division word $\omega = (d_0, \ldots, d_{m-1})$ encoding the m iterates of f of a path, define its *total division count* by

$$D(\omega) := \sum_{j=0}^{m-1} d_j,$$

where d_j equals the 2-adic valuation occurring at the j-th iterate for any trajectory realizing ω.

Thus $D(\omega)$ is precisely the exponent of 2 appearing in the denominator of the linear-affine form of the m-th iterate of f (see Equation (3.1)).

Division words with the same length m may differ in the distribution of divisions by two between steps and hence in their total division count. However, for each division word, its total division count $D(\omega)$ is uniquely determined by the sequence of divisions encoded in the word.

Definition 3.3 (Minimal Total Division Count)**.** Let $A020914(m)$ denote the term of the OEIS sequence A020914, which is the minimal total number of divisions by two among all *admissible* division words containing exactly m iterates of f.

This quantity is given by $\lceil m \log_2(3) \rceil$, a fact derived in Section 4 and tabulated in A020914. Equivalently, it is the smallest integer D such that $2^D \geq 3^m$. Intuitively, each odd step multiplies the trajectory by 3, so the cumulative divisions by two must at least balance this growth; hence the minimal total division count corresponds to the smallest exponent of 2 that offsets the factor of 3^m in the affine map for a path with m iterates of f.

3.4 Density of Residue Classes

Lemma 2 (Disjointness of Residue Classes)**.** Fix $m \geq 0$. The residue classes induced by distinct division words $\omega = (d_0, \ldots, d_{m-1})$ with m iterates of f are pairwise disjoint.

Proof. Let ω be a division word with m iterates of f and total division count

$$D(\omega) = \sum_{j=0}^{m-1} d_j.$$

It determines an affine map

$$x \longmapsto \frac{3^m x + c(\omega)}{2^{D(\omega)}}.$$

For the m-th iterate of f to be an integer, x must satisfy the integrality condition

$$3^m x + c(\omega) \equiv 0 \pmod{2^{D(\omega)}}.$$

Since 3^m is odd, it is coprime to $2^{D(\omega)}$. Hence it is invertible modulo $2^{D(\omega)}$; that is, there exists an integer $(3^m)^{-1} \bmod 2^{D(\omega)}$ such that

$$3^m \cdot (3^m)^{-1} \equiv 1 \pmod{2^{D(\omega)}}.$$

Multiplying the congruence by this inverse yields a unique residue class

$$x \equiv r_\omega \pmod{2^{D(\omega)}}.$$

Distinct division words impose distinct sequences of valuation constraints, and hence define distinct congruence conditions modulo powers of two. These determine different congruence conditions of the form

$$3^m x + c(\omega) \equiv 0 \pmod{2^{D(\omega)}}.$$

Since each such congruence has a unique solution modulo $2^{D(\omega)}$, distinct division words define distinct residue classes. □

Proposition 3.1. A residue class induced by a division word ω has natural density

$$2^{-D(\omega)}.$$

In particular, for classes with m iterates of f,

$$\text{density} \leq 2^{-A020914(m)}.$$

Proof. If $m = 0$, then the division word is empty and $D(\omega) = 0$, so the residue class contains all integers and has natural density $1 = 2^0$.

For $m > 0$, each residue class is an arithmetic progression with modulus $2^{D(\omega)}$ and exactly one representative per period. The natural density of such a class is therefore $2^{-D(\omega)}$. □

Remark. For fixed m, there are finitely many admissible division words with m iterates of f. Let $\widetilde{A}(m)$ denote the number of such words that correspond to convergent residue classes. Since the classes are pairwise disjoint, their total density contribution at iterate depth m is bounded by

$$\widetilde{A}(m)\, 2^{-A020914(m)}.$$

3.5 Example

Consider convergent segments with $m = 4$ iterates of f. The minimal total division count for $m = 4$ is

$$A020914(4) = 7.$$

One admissible division word attaining this minimum is

$$\omega = (1, 1, 1, 4),$$

for which

$$D(\omega) = 1 + 1 + 1 + 4 = 7.$$

Composing the accelerated map symbolically yields

$$f_\omega(x) = \frac{3^4 x + c(\omega)}{2^7} = \frac{81x + 65}{128},$$

so the canonical affine constant for this word is

$$c(\omega) = 65.$$

The integrality condition

$$81x + 65 \equiv 0 \pmod{128}$$

determines a unique residue class

$$x \equiv 15 \pmod{128}.$$

Hence all integers in this class have natural density 2^{-7}.

Interpretation. The word $\omega = (1, 1, 1, 4)$ means:

- At each of the first three iterates of f, $3x + 1$ contains exactly one factor of 2.
- At the fourth iterate of f, $3x + 1$ contains four factors of 2.

Under the accelerated map

$$f(x) = \frac{3x + 1}{2^{v_2(3x+1)}},$$

all of these divisions occur within a single iterate. Thus $d_3 = 4$ means that the fourth iterate divides by 2^4 in that step. The constant $c(\omega)$ is fixed entirely by this symbolic structure and ensures that precisely the integers in the residue class $x \equiv 15 \pmod{128}$ produce integer iterates under the affine form.

This example illustrates that convergence properties depend only on the division word ω and its total division count $D(\omega)$, not on the specific starting value.

4 Grammar of Collatz Division Words

4.1 Division Words as Dynamical Encodings

Remark (Division Grammar as a Dynamical Encoding)**.** Recall from Section 3 that a *division word* $\omega = (d_0, \ldots, d_{m-1})$ records the exact 2-adic valuation $d_i = v_2(3x_i+1)$, occurring at each iterate of the fully accelerated Collatz map.

The division-word grammar introduced in this section provides a symbolic encoding of accelerated Collatz dynamics, following the symbolic approach of Chamberland [8]. This encoding is not injective: multiple positive integers can correspond to the same division word. Specifically, distinct integers whose trajectories share the same sequence of division counts after m iterates of the accelerated map f trace the same division word, forming the arithmetic residue classes described in Section 3.

Unlike a purely combinatorial encoding, division words retain essential dynamical information. Each division word $\omega = (d_0, \ldots, d_{m-1})$ determines the sequence of multiplicative factors of the linear coefficient

$$r_i = \frac{3}{2^{d_i}},$$

so that the contractive or expansive contribution of each iterate is preserved in the linear coefficient (with the affine term handled separately via $c(\omega)$). This structure will allow us to quantify densities of residue classes and to formulate sufficient conditions for descent, even though individual trajectories are collapsed under the encoding.

4.2 Affine Structure of Division Words

Lemma 3 (Explicit Form of the Affine Constant)**.** Let $\omega = (d_0, \ldots, d_{m-1})$ be an admissible division word. Then the affine map

$$F_\omega(x) = \frac{3^m x + c(\omega)}{2^{D(\omega)}}$$

has the explicit form

$$c(\omega) = \sum_{j=0}^{m-1} 3^{m-1-j} \cdot 2^{\sum_{i=0}^{j-1} d_i},$$

where empty sums are defined as 0.

Lemma 4 (Injectivity of the Affine Encoding at Fixed Length)**.** Let $\omega = (d_0, \dots, d_{m-1})$ and $\omega' = (d'_0, \dots, d'_{m-1})$ be admissible division words of the same length m. Let

$$c(\omega) = \sum_{j=0}^{m-1} 3^{m-1-j} \cdot 2^{D_j}, \quad D_j = \sum_{i=0}^{j-1} d_i,$$

and define $c(\omega')$ analogously.

If $\omega \neq \omega'$, then

$$c(\omega) \neq c(\omega').$$

Equivalently, the map $\omega \mapsto c(\omega)$ is injective on admissible division words of fixed length m.

Proof. Suppose $\omega \neq \omega'$, and let

$$k = \min\{\, j : d_j \neq d'_j \,\}$$

be the first index at which the two words differ. Then for all $j < k$, we have $D_j = D'_j$, while $D_k \neq D'_k$.

Consider the difference

$$\Delta := c(\omega) - c(\omega') = \sum_{j=0}^{m-1} 3^{m-1-j}\big(2^{D_j} - 2^{D'_j}\big).$$

The terms with $j < k$ cancel, so

$$\Delta = 3^{m-1-k}\big(2^{D_k} - 2^{D'_k}\big) + \sum_{j=k+1}^{m-1} 3^{m-1-j}\big(2^{D_j} - 2^{D'_j}\big).$$

Let $\delta_k = \min(D_k, D'_k)$. Then

$$2^{D_k} - 2^{D'_k} = 2^{\delta_k}(2^a - 2^b),$$

where one of a, b is zero and $2^a - 2^b$ is odd. Hence

$$v_2\Big(3^{m-1-k}\big(2^{D_k} - 2^{D'_k}\big)\Big) = \delta_k,$$

since 3^{m-1-k} is odd.

For $j > k$, both D_j and D'_j strictly exceed δ_k, since the prefix sums increase by positive integers. Therefore

$$v_2\Big(3^{m-1-j}\big(2^{D_j} - 2^{D'_j}\big)\Big) > \delta_k \quad \text{for all } j > k.$$

It follows that Δ is a sum of one term with 2-adic valuation exactly δ_k and remaining terms of strictly higher 2-adic valuation. Hence

$$v_2(\Delta) = \delta_k,$$

and in particular $\Delta \neq 0$.

Therefore $c(\omega) \neq c(\omega')$, completing the proof. □

Remark (Uniqueness of Encoding)**.** By Lemma 4, the affine constant $c(\omega)$ uniquely determines the division word ω among admissible words of fixed length.

Theorem 5 (Stepwise Realizability of Division Words)**.** Let $\omega = (d_0, \dots, d_{m-1})$ be a division word, and define $c(\omega)$ by

$$c(\omega) = \sum_{j=0}^{m-1} 3^{m-1-j} \cdot 2^{\sum_{i=0}^{j-1} d_i}.$$

Then the identity

$$2^{D(\omega)} x_m = 3^m x_0 + c(\omega)$$

holds if and only if the sequence $(x_0, \dots, x_m)$ satisfies the accelerated Collatz relations

$$x_{i+1} = \frac{3x_i + 1}{2^{d_i}}, \quad 0 \le i < m,$$

with each intermediate step integral.

In particular, the congruence

$$3^m x_0 + c(\omega) \equiv 0 \pmod{2^{D(\omega)}}$$

is equivalent to the existence of a trajectory realizing ω.

Remark (Structural Interpretation)**.** Each term in $c(\omega)$ corresponds to the additive contribution of a single "+1" in the Collatz iteration, propagated forward by subsequent multiplications by 3 and backward-scaled by preceding divisions by 2.

Theorem 6 (Congruence Stability of Division Words)**.** Let ω be admissible and let r_ω satisfy

$$r_\omega \equiv -c(\omega) \cdot (3^m)^{-1} \pmod{2^{D(\omega)}}.$$

Then for all $x \equiv r_\omega \pmod{2^{D(\omega)}}$, the first m iterations of the accelerated Collatz map realize exactly the division word ω.

4.3 Admissibility of Division Words

Definition 4.1 (Admissible Division Word)**.** Let ω be a division word with m iterates of f and total division count $D(\omega)$. Associated to ω is the affine map

$$F_\omega(x) = \frac{3^m x + c(\omega)}{2^{D(\omega)}},$$

where $c(\omega)$ is the canonical integer determined by ω.

The division word ω is said to be *admissible* if there exists at least one integer x such that $F_\omega(x)$ is an integer. Equivalently, ω is admissible if the numerator $3^m x + c(\omega)$ is divisible by $2^{D(\omega)}$ for at least one residue class of x mod $2^{D(\omega)}$.

Remark. Admissibility is therefore an arithmetic condition on the word ω itself, expressed as a divisibility constraint on the associated affine map. When this condition holds, the set of all integers x for which $F_\omega(x)$ is integral forms a unique residue class modulo $2^{D(\omega)}$, corresponding to the residue class induced by ω.

Lemma 7 (Minimal Division Count)**.** For any admissible division word $\omega = (d_0, \dots, d_{m-1})$ of length m,

$$d_0 + \cdots + d_{m-1} \ \geq \ A020914(m),$$

where

$$A020914(m) = \lceil m \log_2(3) \rceil.$$

Equality holds if and only if ω achieves the minimal possible total division count among admissible words of length m.

Definition 4.2 (Convergent Division Word)**.** An admissible division word $\omega = (d_0, \dots, d_{m-1})$ is said to be *convergent* (in the linear sense) if

$$\frac{3^m}{2^{d_0 + \cdots + d_{m-1}}} < 1.$$

Remark (Role of the Affine Term)**.** The constant $c(\omega)$ in the affine representation

$$F_\omega(x) = \frac{3^m x + c(\omega)}{2^{D(\omega)}}$$

encodes the cumulative contribution of the additive term in the classical $3x+1$ map along the trajectory determined by ω.

By the canonical choice of Appendix A, $c(\omega)$ satisfies

$$0 \leq c(\omega) < 2^{D(\omega)},$$

and hence

$$0 \leq \frac{c(\omega)}{2^{D(\omega)}} < 1.$$

Thus F_ω can be written in the form

$$F_\omega(x) = \left(\frac{3^m}{2^{D(\omega)}}\right) x + \varepsilon_\omega, \quad 0 \leq \varepsilon_\omega < 1.$$

Thus the affine term contributes a uniformly bounded offset, while the multiplicative factor governs the asymptotic behavior of the map. In particular, the condition

$$\frac{3^m}{2^{D(\omega)}} < 1$$

ensures that $F_\omega(x) < x$ for all sufficiently large x.

4.4 Residue Classes Induced by Division Words

Fix a division word $\omega = (d_0, \ldots, d_{m-1})$ of length m that is admissible in the sense of Definition 4.1. Let $D(\omega)$ denote its total division count. By admissibility, $D(\omega) \geq A020914(m)$.

Lemma 8 (Residue Class Modulus)**.** Let $x, y \in \mathbb{N}$ both realize the same division word ω, i.e., their trajectories under f exhibit the division counts encoded by ω for the first m iterates of the accelerated map f. Then

$$x \equiv y \pmod{2^{D(\omega)}}.$$

Proof. For any integer z realizing ω, the m iterates of the accelerated map f encoded by ω induce the affine relation

$$F_\omega(z) = \frac{3^m z + c(\omega)}{2^{D(\omega)}}.$$

By admissibility, $F_\omega(z) \in \mathbb{N}$ is integral. This integrality immediately implies that

$$3^m z + c(\omega) \text{ is divisible by } 2^{D(\omega)}.$$

If both x and y realize ω, then

$$3^m x + c(\omega) \equiv 0 \equiv 3^m y + c(\omega) \pmod{2^{D(\omega)}},$$

which implies

$$3^m (x - y) \equiv 0 \pmod{2^{D(\omega)}}.$$

Since 3^m is invertible modulo $2^{D(\omega)}$, it follows that

$$x \equiv y \pmod{2^{D(\omega)}}.$$

$\square$

Lemma 9 (Associated Affine Collatz Map)**.** Each admissible division word ω induces a unique affine map

$$F_\omega(x) = \frac{3^m x + c(\omega)}{2^{D(\omega)}},$$

where $c(\omega)$ is uniquely determined modulo $2^{D(\omega)}$ by the requirement that $F_\omega(x) \in \mathbb{N}$ for all integers x in the residue class induced by ω.

Remark (Affine Compression of Trajectories)**.** For any integer realizing a division word ω, the value $F_\omega(x)$ coincides exactly with the result of applying the accelerated Collatz map through the m iterates of the accelerated map f encoded by ω, including all intermediate divisions by two. Thus F_ω represents a compression of a finite Collatz trajectory into a single affine step.

Theorem 10 (Residue Class Representation of Division Words)**.** Let ω be an admissible division word of length m with total division count $D(\omega)$.

Then there exists a unique residue class

$$r_\omega \pmod{2^{D(\omega)}}$$

such that

$$x \text{ realizes } \omega \iff x \equiv r_\omega \pmod{2^{D(\omega)}}.$$

Equivalently, the set of all integers realizing ω is

$$C_\omega = \{r_\omega + k \cdot 2^{D(\omega)} : k \in \mathbb{N}_0\}.$$

In particular, C_ω is infinite and has density $2^{-D(\omega)}$ within $\mathbb{N}$.

Proof. From the affine representation,

$$F_\omega(x) = \frac{3^m x + c(\omega)}{2^{D(\omega)}},$$

integrality is equivalent to

$$3^m x + c(\omega) \equiv 0 \pmod{2^{D(\omega)}}.$$

Since 3^m is odd, it is invertible modulo $2^{D(\omega)}$, yielding

$$x \equiv -c(\omega) \cdot (3^m)^{-1} \pmod{2^{D(\omega)}}.$$

This determines a unique residue class.

Conversely, any x in this class satisfies the integrality condition, and therefore realizes the division word ω.

Thus admissible division words are in one-to-one correspondence with residue classes modulo powers of two. When viewed from a density perspective, these residue classes will be referred to as *cylinders*. □

Remark (Non-emptiness is constructive)**.** The residue representative r_ω can be computed explicitly via

$$r_\omega \equiv -c(\omega) \cdot (3^m)^{-1} \pmod{2^{D(\omega)}},$$

which provides a direct construction of integers realizing ω.

Remark (Uniform Spacing of Realizations)**.** All integers realizing a division word ω are separated by exactly

$$2^{D(\omega)}.$$

This explains the regular spacing observed in computational data and the repetition of division words across intervals of length $3 \cdot 2^n$.

Remark (Enumeration of Admissible Words)**.** The number of distinct admissible division words with fixed total division count corresponds to A186009. Each such word defines a distinct cylinder of density $2^{-D(\omega)}$, and these cylinders are disjoint.

Lemma 11 (Convergent Residue Class)**.** Each convergent division word ω induces a unique arithmetic residue class

$$x \equiv r_\omega \pmod{2^{D(\omega)}},$$

whose elements all follow the same division pattern for m iterates of the accelerated map f and satisfy

$$F_\omega(x) < x \quad \text{for all} \quad x > \frac{c(\omega)}{2^{D(\omega)} - 3^m}.$$

Consequently, every element of the class reaches a strictly smaller integer after completing the m iterates of the accelerated map f encoded by ω.

Remark (Solving the Affine Inequality)**.** The bound $x > \dfrac{c(\omega)}{2^{D(\omega)} - 3^m}$ is obtained by solving the affine inequality $F_\omega(x) < x$ for x:

$$\frac{3^m x + c(\omega)}{2^{D(\omega)}} < x \implies x > \frac{c(\omega)}{2^{D(\omega)} - 3^m}.$$

This ensures that elements of the residue class indeed descend under the accelerated map.

Proposition 4.1 (Cylinder Density)**.** For any admissible word ω,

$$\mu(C_\omega) = 2^{-D(\omega)}.$$

5 Increment Grammar and Horizontal-Sum Construction

Remark (Increment Grammar as a Counting Mechanism)**.** The first difference sequence of A020914, given by A022921, induces a binary increment grammar $\{s, d\}$ that governs the growth of the minimal admissible total division count.

This grammar reflects the minimal admissibility constraints on cumulative 2-adic valuations, as encoded by A020914, and thus governs the structure of admissible division words at the level of total division count.

This grammar does not encode the division word entries themselves, but instead controls the combinatorial structure used to enumerate admissible division words of fixed length.

In particular, it provides a mechanism for estimating the growth of the counting function $\widetilde{A}(m)$, which will be used to bound the total density of associated residue classes.

Proposition 5.1. The number of convergent division words of length m is given by A186009. This sequence therefore describes the combinatorial growth rate of convergent division words at depth m, and hence of the associated residue classes.

Remark (Canonical Tabular Realization)**.** A canonical realization of this counting sequence via the horizontal-sum construction, together with its derivation from the increment grammar induced by A022921, is given in Appendix E. The appendix serves as a structural reference: the arguments in this section rely only on the grammar itself and not on any numerical properties of the tabulation.

Lemma 12 (Controlled Growth of Horizontal Sums)**.** Let a row have strictly positive entries with total sum S.

A single increment step produces a new row whose total sum is S.

If a double increment is applied immediately afterward, producing a row in which the terminal entry equals the sum of the previous row and all other entries remain strictly positive, then the total sum of the resulting row satisfies

$$S_{\text{new}} < 3S.$$

Proof. The total sum of the original row is S.

Under a double increment step, the resulting row consists of:

1. (i) interior entries with total $S_{\text{int}} < S$, and
2. (ii) two terminal contributions, each equal to S.

Hence the total sum satisfies

$$S_{\text{new}} = S_{\text{int}} + 2S < 3S.$$

□

Remark (Implication for Density Bounds)**.** The lemma shows that each local doubling step increases horizontal sums by at most a fixed multiplicative factor.

Consequently, the number of admissible configurations grows at most exponentially with a controlled base determined by the increment grammar.

When combined with the exponential decay

$$2^{-D(\omega)} \sim 3^{-m}$$

from Section 3, this establishes a direct competition between combinatorial growth and exponential measure decay, with both processes governed by the same base 3.

This balance will be used in the density section to show that the total density contribution of long words vanishes.

6 Finite-State Linear Structure of the Horizontal-Sum Dynamics

This section formalizes the horizontal-sum construction as a collection of positive linear operators acting on finite-dimensional cones. The purpose is to obtain uniform upper bounds on row growth by exploiting monotonicity and positivity.

6.1 Linear Operator Formulation

Let $v_n = (a_1(n), a_2(n), \ldots, a_{k_n}(n))$ denote the generating row at level n, where the length k_n depends on the increment history. Let

$$S_n = \sum_i a_i(n)$$

be its total sum. Ignoring terminal duplication for the moment, the recurrence

$$a_i(n) = a_i(n-1) + a_{i-1}(n)$$

(with the convention that $a_0(n) = 0$) is linear with nonnegative coefficients.

A single increment step therefore corresponds to multiplication by a linear operator T_s with nonnegative entries. A double step duplicates the terminal entry after this linear evolution. This operation is also linear and can be represented by another operator T_d whose matrix entries are again nonnegative.

A single increment acts as a positive linear endomorphism $T_s^{(k)} : \mathbb{R}^k \to \mathbb{R}^k$, preserving row length, whereas a double increment acts as a positive linear map $T_d^{(k)} : \mathbb{R}^k \to \mathbb{R}^{k+1}$, increasing the row length by one via terminal duplication.

Hence each increment symbol $\sigma_n \in \{s, d\}$ induces a positive linear map

$$v_{n+1} = T_{\sigma_n} v_n,$$

and any admissible increment word corresponds to a product of matrices

$$v_{n+k} = T_{\sigma_{n+k-1}} \cdots T_{\sigma_n} v_n,$$

all of whose entries are nonnegative.

6.2 Monotone Extremal Propagation

Lemma 13 (Monotone Extremal Propagation)**.** Let v, w be two admissible rows of equal length satisfying $v_i \le w_i$ for all i. Then for any admissible increment sequence,

$$T_{\sigma_{n+k-1}} \cdots T_{\sigma_n} v \;\le\; T_{\sigma_{n+k-1}} \cdots T_{\sigma_n} w \quad \text{coordinatewise,}$$

and consequently

$$\sum_i (Tv)_i \le \sum_i (Tw)_i.$$

Proof. Each operator T_s and T_d has only nonnegative matrix entries. Thus if $v \le w$ coordinatewise, then $T_s v \le T_s w$ and $T_d v \le T_d w$. Induction over the length of the increment word gives the result. □

This lemma shows that the horizontal-sum dynamics is *order-preserving*. Rows that maximize growth at one stage continue to dominate all others under every admissible future evolution.

6.3 Reduction to Local Pattern Bounds

Let $P = (\sigma_1, \dots, \sigma_k)$ be an admissible local pattern of length k, with associated operator product

$$T_P = T_{\sigma_k} \cdots T_{\sigma_1}.$$

Define

$$\alpha(P) = \sup_{v \ne 0} \frac{\sum_{i=1}^{k_P} (T_P v)_i}{\sum_{i=1}^{k_0} v_i},$$

where k_0 and k_P are the row lengths before and after applying T_P, respectively.

Remark (Why extremal rows suffice)**.** The *positive cone* in $\mathbb{R}^m$ is

$$\mathbb{R}^m_{\ge 0} = \{v \in \mathbb{R}^m : v_i \ge 0\}.$$

The operators T_s and T_d are linear, homogeneous, and have only nonnegative entries. Hence for any pattern P, the operator T_P maps the positive cone into itself and is monotone with respect to the coordinatewise order.

Remark (Reduction to coordinate extremizers)**.** Since T_P is a linear operator with nonnegative entries, the functional

$$v \mapsto \sum_i (T_P v)_i$$

is linear and monotone on the positive cone.

Hence its maximum over the simplex is attained at a coordinate vector, and the supremum equals the maximal column sum of T_P.

By positivity and Lemma 13, the supremum is attained by rows that saturate the inequalities in Lemma 12.

Since admissible increment patterns are precisely those generated by the increment grammar, these bounds apply uniformly across all admissible division words.

Hence the extremal examples constructed in Appendix D provide valid upper bounds for $\alpha(P)$ for every admissible occurrence of P.

6.4 Maximal Growth via Grammar Cycles

Any admissible infinite increment sequence decomposes into a concatenation of finite grammar-allowed blocks.

The growth of each block is bounded by its corresponding factor $\alpha(P)$. Therefore the long-term growth is bounded by products over these finite patterns.

It follows that the asymptotic growth rate is controlled by maximal cycle products:

$$\limsup_{n\to\infty} S_n^{1/n} \le \max_{\text{admissible cycles } C} \left(\prod_{P\in C} \alpha(P) \right)^{1/|C|}.$$

This reduces global growth control to the finite enumeration of local patterns performed in Appendix D, leading to the bound $\rho_5 < 3$ used in Theorem 16.

7 Cumulative Density of Convergent Classes

Remark (Roadmap)**.** In this section we combine the constructions developed in Sections 4 and 5 to analyze the natural density of convergent residue classes. First, we record the number of admissible residue classes of stopping time i via the sequence $\widetilde{A}(i) = A186009(i+1)$ and the associated minimal denominator growth $B(i) = A020914(i)$. Then, using uniform bounds on $2^{B(i)}$ and the grammar-enforced suppression of locally expansive horizontal-sum growth, we show that the per-step growth of $\widetilde{A}(i)$ is strictly dominated by the denominator. Finally, this establishes that the density contribution of newly appearing convergent residue classes vanishes as $i \to \infty$, implying that almost every positive integer belongs to at least one convergent residue class.

Remark (Cylinders as Measure Representatives)**.** By Theorem 10, each admissible division word ω corresponds to a residue class (cylinder)

$$C_\omega = \{x \equiv r_\omega \pmod{2^{D(\omega)}}\}$$

of natural density $2^{-D(\omega)}$.

In this section, cylinders serve only as a convenient representation of subsets of $\mathbb{N}$ with explicitly computable density. All structural and dynamical properties are determined by the admissible division words and the increment grammar; cylinders enter only in the final measure-theoretic accounting.

7.1 Density Contributions and Denominator Growth

Define

$$\widetilde{A}(i) := A186009(i+1),$$

so that $\widetilde{A}(i)$ counts convergent residue classes of stopping time i.

Let

$$B(i) := A020914(i),$$

Thus the density contribution of convergent residue classes of stopping time i is the total measure of the corresponding cylinders:

$$\sum_{\omega\colon |\omega|=i} 2^{-D(\omega)}.$$

That is, density is computed by summing the measures of the cylinders corresponding to admissible words of length i.

For minimal stopping time classes, all such words satisfy $D(\omega) = B(i)$, and hence this reduces to

$$\widetilde{A}(i)\, 2^{-B(i)}.$$

By definition of A020914 as the minimal exponent with $2^{B(i)} > 3^i$, the function $B(i)$ satisfies the following properties:

- $B(i)$ is strictly increasing, with increments either 1 or 2 (never zero).
- By construction, $2^{B(i)} > 3^i$ for all $i \geq 0$.
- Consecutive powers of two differ by a factor of 2, so

$$3^i < 2^{B(i)} < 2 \cdot 3^i.$$

Because $B(i)$ is defined as the minimal integer with $2^{B(i)} > 3^i$, the quantity $2^{B(i)}$ is the smallest power of two exceeding 3^i.

Constant-factor control. This bound depends only on the definition of A020914 and does not rely on any delayed-doubling or numerator-growth arguments. Consequently, the reciprocal satisfies

$$2^{-B(i)} \in \left(\frac{1}{2 \cdot 3^i}, \frac{1}{3^i} \right),$$

providing a *uniform constant-factor bound* for the density contribution at stopping time i.

We now record this observation formally, as it provides the fixed exponential scale against which numerator growth will be compared.

Lemma 14 (Uniform Control of Density Contributions)**.** For all $i \geq 0$,

$$3^i < 2^{A020914(i)} < 2 \cdot 3^i.$$

Consequently, the density contribution of convergent residue classes of stopping time i, given by $\widetilde{A}(i) 2^{-A020914(i)}$, satisfies

$$\frac{\widetilde{A}(i)}{2 \cdot 3^i} \; < \; \widetilde{A}(i)\, 2^{-A020914(i)} \; < \; \frac{\widetilde{A}(i)}{3^i}.$$

Proof. By definition, $A020914(i)$ is the minimal integer m such that $2^m > 3^i$. Consecutive powers of two differ by a factor of 2, so

$$3^i < 2^{A020914(i)} < 2 \cdot 3^i.$$

Taking reciprocals and multiplying by $\widetilde{A}(i)$ gives the stated bounds. $\square$

Lemma 15 (Disjointness of Minimal Convergent Classes)**.** For each $i \geq 0$, the residue classes counted by

$$\widetilde{A}(i) = A186009(i+1)$$

correspond to congruence classes of *minimal stopping time* i and are pairwise disjoint.

Distinct admissible division words of length i yield distinct cylinders modulo $2^{B(i)}$ (Theorem 10). Since these cylinders are disjoint residue classes, their associated sets of integers are disjoint.

Minimality of the stopping time ensures that no integer in such a cylinder belongs to a cylinder of shorter length, completing the argument.

Proof. Each admissible division word of length i determines a residue class modulo $2^{B(i)}$ consisting of integers whose first descent below their initial value occurs precisely at step i. Minimality of the stopping time excludes the possibility that such an integer satisfies a shorter admissible division word. Distinct admissible division words of length i yield distinct residue classes modulo $2^{B(i)}$, and therefore their associated sets of integers are disjoint. Since each class is realized modulo the same modulus $2^{B(i)}$, disjointness as congruence classes implies disjointness as subsets of $\mathbb{N}$. $\square$

7.2 Growth Rate of the Number of Convergent Classes

Remark (Grammar-Enforced Suppression of Expansive Growth)**.** The proof of Theorem 16 does not rely on the absence of locally expansive configurations. Certain increment patterns are expansive in isolation, but the increment grammar forbids them or requires surrounding contractive structure.

(1) Local expansivity versus admissibility. A single double increment $\{d\}$ or the short pattern $\{s, d, d\}$ produces horizontal-sum growth with geometric mean exceeding 1. However, neither pattern is admissible under the increment grammar induced by A022921, as all admissible sequences must begin with $\{s\}$ and no occurrence of $\{d, d\}$ may appear without sufficient separating structure.

(2) Minimal realization of expansion. The shortest admissible pattern containing a double increment dd is the length-5 block

$$\{s, d, s, d, d\},$$

which constitutes the minimal expansive configuration. All admissible patterns of length ≤ 4 are contractive.

(3) Forced contractive buffering. The grammar enforces that the expansive 5-cycle cannot appear in isolation. The minimal admissible prefix leading to a $\{d, d\}$ occurrence is the length-7 block

$$\{s, d, s, d, s, d, d\},$$

which is contractive. Concatenating the expansive 5-cycle with this prefix yields a length-12 block that is also contractive. Thus expansive behavior is always dominated by mandatory contractive buffers.

(4) Consequence for asymptotic growth. Any admissible infinite increment sequence decomposes into finite grammar-allowed blocks. Horizontal-sum growth is submultiplicative under concatenation, so the asymptotic per-step growth rate is bounded above by the largest geometric mean among admissible blocks. Expansive blocks are separated by uniformly bounded contractive buffers, ensuring their asymptotic density in any admissible sequence is strictly less than one.

Theorem 16 (Maximal Admissible Growth Rate)**.** Let $\widetilde{A}(i) = A186009(i+1)$ denote the number of convergent residue classes of stopping time i, generated by the horizontal-sum construction associated with the increment grammar of A022921. Let $B(i) := A020914(i)$ denote the cumulative exponent of 2 up to level i.

The per-step growth of $\widetilde{A}(i)$ is bounded by the maximal geometric growth rate of a 5–cycle (see Appendix D, Equation (D.2)):

$$\limsup_{i\to\infty} \widetilde{A}(i)^{1/i} \leq \rho_5 < 3,$$

and consequently

$$\frac{\widetilde{A}(i)}{2^{B(i)}} \longrightarrow 0 \quad \text{as } i \to \infty.$$

Proof. The increment sequence A022921 induces a finite symbolic grammar on $\{s, d\}$, corresponding to single and double increments in the horizontal-sum construction.

Among all admissible sequences, the shortest potentially expansive configuration is the *5-cycle*

$$\{s, d, s, d, d\}.$$

All admissible words of length ≤ 4 produce horizontal-sum growth factors with geometric mean ≤ 1. A complete enumeration of admissible local patterns and their growth factors is given in Appendix D.

Explicit computation of the net horizontal-sum growth factor for the 5-cycle gives

$$\rho_5 := \big(\text{horizontal sum multiplier over the 5-cycle}\big)^{1/5} < 3.$$

Any admissible infinite increment sequence decomposes into concatenations of finite blocks permitted by the grammar. As explained in Remark 7.2, all locally expansive blocks are separated by mandatory contractive buffering. Horizontal-sum growth is submultiplicative under concatenation, so the asymptotic per-step geometric growth is bounded above by the maximal geometric mean among admissible blocks, which is attained by the 5-cycle and denoted ρ_5. Hence

$$\limsup_{i\to\infty} \widetilde{A}(i)^{1/i} \leq \rho_5 < 3.$$

From Lemma 14, the denominator satisfies

$$3^i < 2^{B(i)} < 2 \cdot 3^i.$$

Consequently, there exists a constant $C > 0$, depending only on initial conditions and finite prefixes, such that

$$\widetilde{A}(i) \leq C\,\rho_5^i \quad \text{for all } i.$$

Dividing by $2^{B(i)}$ gives

$$\frac{\widetilde{A}(i)}{2^{B(i)}} \leq \frac{C\,\rho_5^i}{3^i} \longrightarrow 0,$$

which completes the proof. $\square$

7.3 Vanishing of the Cumulative Density

The preceding results show that although the number of convergent residue classes $\widetilde{A}(i)$ grows with stopping time, its growth rate is strictly dominated by the denominator $2^{B(i)}$, which grows asymptotically like 3^i.

By Lemma 14, the density contribution at level i satisfies

$$\frac{\widetilde{A}(i)}{2 \cdot 3^i} < \widetilde{A}(i)\, 2^{-B(i)} < \frac{\widetilde{A}(i)}{3^i}.$$

By Theorem 16, we have

$$\frac{\widetilde{A}(i)}{3^i} \longrightarrow 0 \quad \text{as } i \to \infty.$$

Although the per-level density contribution already tends to zero, the stronger geometric bound proved in Theorem 16 implies summability of the contributions, which is the key fact for the cumulative density argument below.

Lemma 17 (Summability of Density Contributions)**.** The series

$$\sum_{i=0}^{\infty} \widetilde{A}(i)\, 2^{-B(i)}$$

converges.

Proof. By Theorem 16, there exists $C > 0$ such that

$$\widetilde{A}(i) \le C\, \rho_5^i,$$

with $\rho_5 < 3$. By Lemma 14,

$$2^{B(i)} > 3^i.$$

Hence

$$\widetilde{A}(i)\, 2^{-B(i)} \le C \left(\frac{\rho_5}{3}\right)^i.$$

Since $\rho_5/3 < 1$, the right-hand side is a geometric sequence with ratio less than 1, and therefore the series converges. □

Lemma 18 (Vanishing Tail Implies Full Density)**.** Let $\{E_i\}$ be a disjoint family of subsets of $\mathbb{N}$ arising from minimal stopping-time classes, and suppose

$$\sum_{i=0}^{\infty} \mu(E_i) < \infty.$$

If the tail sums satisfy

$$\lim_{N \to \infty} \sum_{i \ge N} \mu(E_i) = 0,$$

then the complement of $\bigcup_i E_i$ has natural density zero.

Because the density contributions form a convergent series (Lemma 17), the tail of the series can be made arbitrarily small. Hence for any $\epsilon > 0$, there exists i_0 such that

$$\sum_{i \geq i_0} \widetilde{A}(i)\, 2^{-B(i)} < \epsilon.$$

Thus the hypotheses of Lemma 18 are satisfied, and the complement of the union of convergent cylinders has natural density zero.

In other words, the set of positive integers that fail to belong to any convergent cylinder has natural density zero. By the descent properties established in Section 4, membership in such a class implies eventual decrease under the map.

Thus the family of convergent cylinders forms a disjoint collection whose total measure is finite:

$$\sum_{i=0}^{\infty} \widetilde{A}(i)\, 2^{-B(i)} < \infty.$$

By Lemma 17, the total measure of the disjoint family of convergent cylinders is finite. Applying Lemma 18, it follows that the complement — the set of integers that do not lie in any convergent cylinder — has natural density zero.

Equivalently, the union of all convergent cylinders has density one.

Remark on growth bounds. The preceding argument is purely measure-theoretic and does not rely on lower bounds for the growth of $\widetilde{A}(i)$. Nevertheless, it is possible to establish a nontrivial exponential lower bound on the growth rate of $\widetilde{A}(i)$ by comparison with the auxiliary sequence A047749, which admits a closed-form binomial representation. This comparison shows that the exponential growth rate of $\widetilde{A}(i)$ is bounded below by $\sqrt{27/4}$. The derivation of this bound is independent of the density argument and is developed separately in Appendix G.

We now reinterpret this full-density result in terms of descent dynamics.

8 Global Descent in Density

We now translate the density-one result of Section 7 into a dynamical statement about descent under the accelerated Collatz map.

8.1 Descent Words and Partition of the Descending Set

Theorem 19 (Natural Global Descent)**.** The union of all convergent cylinders has natural density one. Each convergent cylinder consists entirely of integers that admit a finite descent time under the accelerated Collatz map. Consequently, almost every positive integer eventually reaches a smaller value under iteration of the accelerated Collatz map, in the sense of natural density.

Proof. By Section 7, the complement of the union of all convergent cylinders has natural density zero.

Each convergent cylinder consists entirely of integers that admit a finite descent time under the accelerated map. Therefore, for almost every positive integer x (in the sense of natural density), there exists a finite m such that

$$f^m(x) < x.$$

□

Remark (Cylindrical identification)**.** We identify each residue class R_ω with its corresponding cylinder C_ω, i.e. $C_\omega := R_\omega$.

Remark (Minimal Descent Indexing)**.** Minimal descent is defined with respect to m, i.e., the first time an accelerated iterate satisfies $f^m(x) < x$. All cylinders C_ω in this section are indexed by m, so summations over $\mathcal{W}_{\min}$ correctly account for density.

Definition 8.1 (Descent Word)**.** An admissible division word ω of length m is called a *descent word* if for all integers x in its associated cylinder C_ω,

$$F_\omega(x) < x.$$

Definition 8.2 (Minimal Descent Word)**.** A descent word ω is *minimal* if no proper prefix of ω is itself a descent word.

Lemma 20 (Prefix Descent Sufficiency)**.** If ω is a descent word and η is any extension of ω (i.e. $\eta = \omega \star \tau$ for some nonempty suffix τ), then the cylinder C_η is a subset of

C_ω. Consequently, all integers realizing η have already descended at the prefix stage ω.

Here, $|\omega| = m$ counts the number of odd steps in the accelerated map f.

Proof. Any integer x realizing η realizes ω during its first $|\omega| = m$ accelerated iterations. Thus $x \in C_\omega$, so $C_\eta \subseteq C_\omega$. Since ω is a descent word, $F_\omega(x) < x$, and descent occurs before the suffix τ is applied. $\square$

Theorem 21 (Minimal Descent Partition)**.** Let $\mathcal{W}_{\min}$ denote the set of minimal descent words. Then:

1. **Uniqueness.** Every integer x that eventually descends (i.e., there exists a finite m such that $f^m(x) < x$ under the accelerated map) admits a unique minimal descent word $\omega \in \mathcal{W}_{\min}$ such that $x \in C_\omega$.
2. **Disjointness.** If $\omega_1, \omega_2 \in \mathcal{W}_{\min}$ with $\omega_1 \neq \omega_2$, then
$$C_{\omega_1} \cap C_{\omega_2} = \varnothing.$$
3. **Exhaustion.** The set of all integers that ever descend equals
$$\bigcup_{\omega \in \mathcal{W}_{\min}} C_\omega.$$

Proof. (1) For any x that descends, consider the first time m such that $f^m(x) < x$. The division word of length m realized by x is a descent word, and no shorter prefix can be, so it is minimal.

(2) If $x \in C_{\omega_1} \cap C_{\omega_2}$, then x realizes both division words, so their realized division sequences agree up to the minimum of their lengths. Minimality forces $\omega_1 = \omega_2$.

(3) Every descending integer has a minimal descent prefix by (1), and Lemma 20 ensures extensions produce no new elements. $\square$

Corollary 22 (No Overcounting of Density)**.** The residue classes C_ω for $\omega \in \mathcal{W}_{\min}$ are pairwise disjoint. Hence
$$\mu\left(\bigcup_{\omega \in \mathcal{W}_{\min}} C_\omega\right) = \sum_{\omega \in \mathcal{W}_{\min}} \mu(C_\omega),$$
and extensions of descent words contribute no additional density.

All minimal descent words are indexed by m, and any extension $\eta = \omega \star \tau$ does not add new integers; thus summing over $\mathcal{W}_{\min}$ correctly accounts for the total density.

8.2 Natural Global Descent of Almost Every Integer

Remark (Exceptional Orbits)**.** This theorem does not preclude the existence of exceptional trajectories outside the convergent residue classes. Any set not covered by the convergent cylinders constructed above has natural density zero, so the result establishes descent in a measure-theoretic sense rather than an absolute guarantee.

9 Hypothetical Zero-Density Trajectories

The density result established above does not exclude the logical possibility of non-convergent Collatz trajectories. However, it severely restricts the form that any such trajectory could take. In particular, any exceptional set must lie entirely outside the union of convergent cylinders whose cumulative natural density approaches one, confining any hypothetical exception to a set of natural density zero.

Consequently, an exceptional trajectory could not belong to any cylinder families constructed in the preceding sections, nor arise from the structured division patterns that generate those classes. Any trajectory avoiding all convergent cylinders would lie outside all density-generating patterns admitted by the increment grammar, and hence outside the symbolic structures responsible for positive-density behavior.

In this sense, any hypothetical exception would be *dynamically isolated within a set of zero natural density.* It would not be supported by a scalable arithmetic structure or by a cylinder hierarchy of positive density, but would instead persist only as a sparse, non-propagating phenomenon within the integer space.

10 Conclusion

In this work, we analyzed the Collatz map via a decomposition of the positive integers into disjoint cylinders determined by finite division words. Each cylinder corresponds to a minimal stopping-time structure, and its density is determined by the associated exponent growth function $B(i)$. By enumerating admissible division words and comparing their combinatorial growth against powers of two, we showed that the cumulative density of all convergent classes is one. Equivalently, the complement of the union of these cylinders has natural density zero.

The analysis is further sharpened by showing that descent behavior admits a canonical minimal description: integers that eventually descend are partitioned into disjoint cylinders indexed by minimal descent words, and longer admissible division patterns introduce no additional density. This reduces the global analysis to a non-redundant symbolic decomposition of all positive-density trajectories.

This density-based approach establishes that almost every positive integer eventually reaches a strictly smaller value under the accelerated Collatz map. While the argument does not preclude zero-density exceptional trajectories, it provides a strong measure-theoretic framework for understanding the typical dynamics of the map.

Viewed abstractly, this perspective reframes the Collatz problem as an analysis of integer lattices partitioned by arithmetic cylinders, highlighting how division- and cylinder-based combinatorial structures govern discrete dynamical systems. It emphasizes the distinction between "almost all" and "all," yielding a credible, rigorous statement fully consistent with the density analysis.

A Canonical Determination of the Constant $c(\omega)$

Each division word ω induces an affine map $F_\omega(x)$ of length m (see Section 3)

$$F_\omega(x) = \frac{3^m x + c(\omega)}{2^{D(\omega)}}, \qquad D(\omega) = d_0 + \cdots + d_{m-1},$$

where $c(\omega)$ is uniquely defined modulo $2^{D(\omega)}$. In this appendix, we describe a canonical choice and a simple method to compute it.

A.1 Canonical Choice

For a fixed ω, any integer x in the corresponding residue class satisfies

$$3^m x + c \equiv 0 \pmod{2^{D(\omega)}}.$$

We remove ambiguity by selecting the unique representative

$$0 \le c(\omega) < 2^{D(\omega)}.$$

This choice depends only on ω and is independent of the representative x.

A.2 Step-by-Step Computation

Given ω of length m and total division count $D(\omega)$:

1. Choose any integer x in the residue class corresponding to ω.
2. Compute $y = 3^m x$.
3. Let $r = \lceil y/2^{D(\omega)} \rceil$ be the smallest integer such that $r \cdot 2^{D(\omega)} \ge y$.
4. Define $c(\omega) = r \cdot 2^{D(\omega)} - y$.

By this construction, $0 \le c(\omega) < 2^{D(\omega)}$ ensuring that

$$F_\omega(x) = \frac{3^m x + c(\omega)}{2^{D(\omega)}} \in \mathbb{Z} \quad \text{for all } x \text{ in the class.}$$

A.3 Example

Let $\omega = (1, 1, 1, 4)$ with $m = 4$ and $D(\omega) = \sum_{i=0}^{3} d_i = 1 + 1 + 1 + 4 = 7$:

1. Choose $x = 15$. Compute $y = 3^4 \cdot 15 = 1215$.
2. $2^7 = 128$, so $r = \lceil 1215/128 \rceil = 10$.
3. $c(\omega) = 10 \cdot 128 - 1215 = 65$.

Hence the canonical map is

$$F_\omega(x) = \frac{81x + 65}{128}.$$

Checking another representative $x = 143$:

$$F_\omega(143) = \frac{81 \cdot 143 + 65}{128} = 91,$$

confirming that the same $c(\omega)$ works for the entire residue class.

Remark

This construction is purely illustrative and is not required for the density or extremal growth arguments in the main text. It demonstrates concretely how the affine map is fully determined by the division word ω.

Since any two integers realizing the same division word differ by a multiple of $2^{D(\omega)}$, the value of $c(\omega)$ computed by this procedure is independent of the chosen representative.

B Reconstruction of Integers from a Division Word

B.1 Inverse Cylinder Construction

Let $\omega = (d_0, \dots, d_{m-1})$ be an admissible division word with total division count

$$D(\omega) = \sum_{i=0}^{m-1} d_i.$$

From Section 4, ω induces the affine map

$$F_\omega(x) = \frac{3^m x + c(\omega)}{2^{D(\omega)}}.$$

Admissibility implies the congruence condition

$$3^m x + c(\omega) \equiv 0 \pmod{2^{D(\omega)}}.$$

Since 3^m is invertible modulo $2^{D(\omega)}$, this yields the unique residue class

$$x \equiv -c(\omega)\,(3^m)^{-1} \pmod{2^{D(\omega)}}.$$

B.2 Canonical Parametrization of the Cylinder

Let r_ω denote the canonical representative of this residue class in $\{0, 1, \dots, 2^{D(\omega)}-1\}$. Then all integers realizing ω are given by

$$x(k) = r_\omega + k \cdot 2^{D(\omega)}, \quad k \in \mathbb{N}_0.$$

B.3 Affine Reconstruction Form

Applying F_ω to $x(k)$ yields

$$F_\omega(x(k)) = \frac{3^m(r_\omega + k2^{D(\omega)}) + c(\omega)}{2^{D(\omega)}}.$$

Expanding,

$$F_\omega(x(k)) = k \cdot 3^m + \frac{3^m r_\omega + c(\omega)}{2^{D(\omega)}}.$$

Define the constant

$$K_\omega := \frac{3^m r_\omega + c(\omega)}{2^{D(\omega)}}.$$

Then the affine form becomes

$$F_\omega(x(k)) = k \cdot 3^m + K_\omega.$$

B.4 Interpretation

Thus each division word induces a bijection between:

- integers in a residue class $x(k)$, and
- affine images of the form $k \cdot 3^m + K_\omega$.

This provides a complete inverse description of the division-word cylinder.

B.5 Example

Solving the congruence

$$3^4 x + 65 \equiv 0 \pmod{128}$$

gives

$$81x + 65 \equiv 0 \pmod{128}.$$

Reducing $81 \equiv -47 \pmod{128}$, we obtain

$$-47x + 65 \equiv 0 \pmod{128} \quad \Longrightarrow \quad 47x \equiv 65 \pmod{128}.$$

Since 47 is invertible modulo 128, with

$$47^{-1} \equiv 79 \pmod{128},$$

we multiply both sides to obtain

$$x \equiv 79 \cdot 65 \equiv 15 \pmod{128}.$$

Thus the division word $\omega = (1, 1, 1, 4)$ generates the residue class

$$x = 15 + 128k, \quad k \in \mathbb{N}_0.$$

B.6 Derivation of the Affine Constant for $m = 4$

Let $\omega = (a, b, c, d)$ be a division word of length $m = 4$, with total division count

$$D(\omega) = a + b + c + d.$$

Define the accelerated Collatz steps:

$$x_1 = \frac{3x_0 + 1}{2^a}, \quad x_2 = \frac{3x_1 + 1}{2^b}, \quad x_3 = \frac{3x_2 + 1}{2^c}, \quad x_4 = \frac{3x_3 + 1}{2^d}.$$

Multiplying each equation by the corresponding power of 2 gives:

$$2^a x_1 = 3x_0 + 1,$$

$$2^b x_2 = 3x_1 + 1,$$
$$2^c x_3 = 3x_2 + 1,$$
$$2^d x_4 = 3x_3 + 1.$$

We now eliminate variables by backward substitution.

From the last equation:

$$2^d x_4 = 3x_3 + 1.$$

Multiply by 2^c:

$$2^{c+d} x_4 = 3(2^c x_3) + 2^c.$$

Substitute $2^c x_3 = 3x_2 + 1$:

$$2^{c+d} x_4 = 3(3x_2 + 1) + 2^c = 3^2 x_2 + 3 + 2^c.$$

Multiply by 2^b:

$$2^{b+c+d} x_4 = 3^2(2^b x_2) + 3 \cdot 2^b + 2^{b+c}.$$

Substitute $2^b x_2 = 3x_1 + 1$:

$$2^{b+c+d} x_4 = 3^2(3x_1 + 1) + 3 \cdot 2^b + 2^{b+c} = 3^3 x_1 + 3^2 + 3 \cdot 2^b + 2^{b+c}.$$

Multiply by 2^a:

$$2^{a+b+c+d}x_4 = 3^3(2^a x_1) + 3^2 \cdot 2^a + 3 \cdot 2^{a+b} + 2^{a+b+c}.$$

Substitute $2^a x_1 = 3x_0 + 1$:

$$2^{D(\omega)}x_4 = 3^3(3x_0 + 1) + 3^2 \cdot 2^a + 3 \cdot 2^{a+b} + 2^{a+b+c}.$$

Simplifying:

$$2^{D(\omega)}x_4 = 3^4 x_0 + 3^3 + 3^2 \cdot 2^a + 3 \cdot 2^{a+b} + 2^{a+b+c}.$$

Thus we obtain the identity:

$$2^{D(\omega)}x_4 = 3^4 x_0 + \big(3^3 + 3^2 \cdot 2^a + 3 \cdot 2^{a+b} + 2^{a+b+c}\big).$$

This yields the explicit form:

$$c(\omega) = 3^3 + 3^2 \cdot 2^a + 3 \cdot 2^{a+b} + 2^{a+b+c}.$$

C Extremal Local Growth Analysis

Lemma 23 (Local Growth Bounds)**.** For each admissible local pattern occurring within A022921, the ratio of the total sum of entries after applying the pattern to the total sum before applying it is bounded above as follows, uniformly over all admissible placements of (delayed) doubling within the pattern:

Pattern	Max Growth Factor
s, d	3
d, d	123/33
s, d, s	341/131
d, d, s	329/123

Here s denotes a single increment and d a double increment in the horizontal-sum construction. Note that these bounds are independent of the starting row and only depend on the local pattern.

Proposition C.1 (Extremal 5-Cycle Growth)**.** The maximal geometric growth rate achievable by any admissible infinite word under the horizontal-sum dynamics associated with A186009 is bounded by

$$\rho_5 = \left(3^2 \cdot (123/33)^1 \cdot (341/131)^1 \cdot (329/123)^1\right)^{1/5} \approx 2.976 < 3,$$

where the exponents record the number of times each admissible local pattern appears within a single admissible 5-cycle (the minimal admissible expansive block). See Section D.1 for explicit enumeration: two occurrences of $\{s, d\}$, one of $\{d, d\}$, one of $\{s, d, s\}$, and one of $\{d, d, s\}$.

This enumeration ensures that every admissible infinite word can be decomposed into concatenations of 5–cycles whose aggregate growth is controlled by ρ_5. See Lemma 24 for a rigorous argument that 5-cycles cannot be avoided and therefore this bound applies globally.

Lemma 24 (Inevitable 5-Cycle Insertion)**.** Every contractive sequence of cycles under the admissibility grammar of A022921 necessarily contains at least one embedded 5-cycle within either a 12-cycle (7-cycle + 5-cycle) or as a standalone 5-cycle following sufficient consecutive 12-cycles.

In particular:

1. Purely consecutive 7-cycles cannot occur.
2. Expansive 5-cycles appear at bounded intervals within any admissible infinite word.
3. After sequences of three or four consecutive 12-cycles (yielding 41- or 53-cycle blocks), the grammar forces a 5-cycle insertion without exceeding the contractive bound of unity.
4. It is therefore impossible to indefinitely avoid 5-cycles; they are structurally unavoidable.

Remark (Extremal 5-Cycle Envelope)**.** Although the 12-cycle is contractive overall, it contains a 5-cycle in its tail. Repeated sequences of 12-cycles may occasionally yield consecutive 5-cycles once enough contractive slack has accumulated, as seen in the 41-cycle ($3 \times 12 + 5$) or 53-cycle ($4 \times 12 + 5$) structures, but the grammar ensures that 5-cycles cannot be skipped indefinitely.

This observation justifies treating an infinite repetition of 5-cycles as a strict upper envelope: it saturates all local growth factors, while any admissible word, including A186009, contains interspersed contractive 12-cycles, so its per-symbol growth is strictly below the 5-cycle maximal rate ρ_5.

Remark (Connection to Theorem 16)**.** The quantity ρ_5 defined above represents the maximal geometric growth factor per symbol for any admissible word under the horizontal-sum dynamics. Taking the fifth root over one full 5-cycle yields the maximal per-symbol growth, which directly controls $\limsup_{i\to\infty} \widetilde{A}(i)^{1/i}$ in Theorem 16. Because $\rho_5 < 3$, the growth of the number of convergent residue classes is asymptotically strictly slower than the corresponding powers of two, guaranteeing that the density of integers with large stopping times vanishes.

Remark (Derivation and Attribution of Local Growth Constants)**.** The horizontal-sum framework used to analyze $\widetilde{A}(i)$ is inspired by the Pascal-type summation approach introduced by Winkler [9], which provided the original insight that admissible Collatz division words can be organized into structured combinatorial growth processes.

The present work adapts this perspective to the specific setting of A022921 and A186009, with one key modification: unlike Winkler's vertical-sum formulation, the horizontal orientation used here is essential for preserving row-wise combinatorial structure, enabling precise extremal growth calculations. This extension allows a complete analysis of local growth configurations and the derivation of the uniform 5-cycle bound.

The constants shown in Lemma 23 are obtained by examining the horizontal-sum dynamics governing $\widetilde{A}(i)$. Within each local configuration, elements arising from single steps evolve according to standard Pascal-type summation, yielding maximal growth proportional to powers of 2. Additional entries introduced by doubling steps are bounded by summing on top of this baseline growth.

For each admissible local pattern occurring within A022921, all possible placements of delayed doubling contributions were examined, and the largest achievable ratio of post-pattern to pre-pattern row sums was recorded. No assumptions beyond the combinatorial structure of the horizontal-sum construction are used.

A complete derivation of the horizontal-sum mechanism, together with detailed computations of all local growth constants and their admissible concatenations, is given in Appendix D. Consequently, while the conceptual framework traces back to Winkler's original insight, the bounds employed here are established entirely within the present paper. Note that no analytic or probabilistic assumptions are made; all bounds arise from finite combinatorial enumeration.

It follows that any admissible sequence under A022921, including those relevant for A186009, is strictly dominated in per-symbol growth by a concatenation of 5-cycles, yielding the uniform bound of Proposition C.1.

D Extremal Local Growth Calculations

This appendix provides the explicit finite computations underlying the local growth bounds used in the density argument. Recall that the sequence A020914 has first differences given by A022921, consisting of repeated blocks of *single* steps (s) and *double* steps (d).

D.1 Cycle Structure and Net Multiplicative Effect

A frequently occurring block is the 7–cycle

$$\{s, d, s, d, s, d, d\},$$

which contains 3 singles and 4 doubles. The corresponding number of factors of 2 is

$$3 + 2(4) = 11,$$

so the net multiplicative contribution of this cycle satisfies

$$\frac{2^{11}}{3^7} < 1.$$

Thus the 7–cycle is strictly contractive. Every 7–cycle is followed by a 5–cycle

$$\{s, d, s, d, d\},$$

which contains 2 singles and 3 doubles, contributing

$$2 + 2(3) = 8$$

factors of 2. Since

$$\frac{2^8}{3^5} > 1,$$

the 5–cycle is expansive.

Concatenating the 7–cycle and 5–cycle produces a 12–cycle

$$\{s, d, s, d, s, d, d, s, d, s, d, d\},$$

containing 5 singles and 7 doubles. The total number of factors of 2 is

$$5 + 2(7) = 19,$$

and hence

$$\frac{2^{19}}{3^{12}} < 1.$$

Therefore, the 12–cycle is again contractive.

Repeated 12-cycles remain contractive, suppressing growth until enough slack accumulates to allow a 5-cycle insertion. The 5-cycle represents the unique locally expansive pattern, though its repetition is constrained by the grammar.

To visualize the interaction between contractive and expansive cycles, Figure 1 depicts the state transitions of A022921. The red edges indicate the locally expansive 5-cycle, the blue edges the contractive 7-cycle, and purple edges highlight shared transitions.

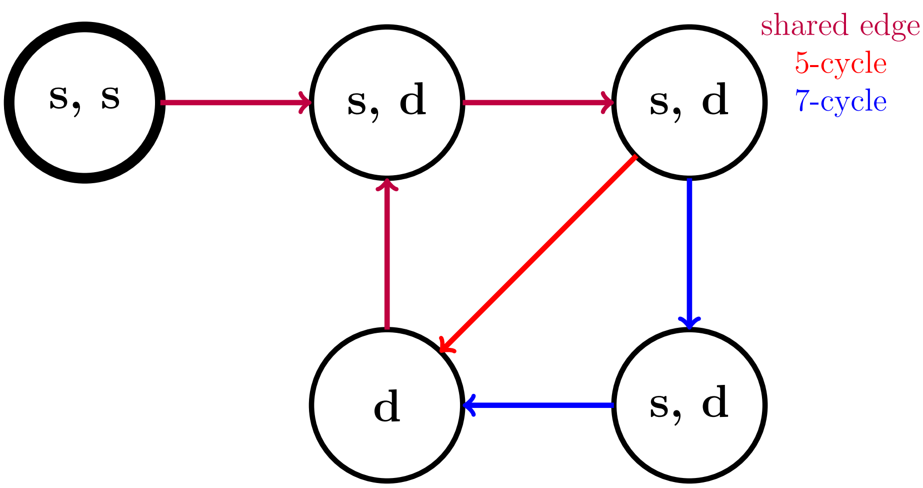

Figure 1: State Transition Diagram for A022921 highlighting cycles: contractive 7-cycle (blue), locally expansive 5-cycle (red), shared edges (purple). Lowercase letters reflect single (s) and double (d) division steps consistent with the main text notation.

The figure highlights the locally expansive 5-cycle (red), the contractive 7-cycle (blue), and the shared edges (purple), visually illustrating the grammar constraints of the sequence A022921.

Horizontal-Sum Dynamics and Pascal-Type Growth

The generating rows evolve by a horizontal-sum rule:

$$a_i(n) = a_i(n-1) + a_{i-1}(n), \quad i, n \in \mathbb{N}, \tag{D.1}$$

so that sequences of single steps exhibit ordinary Pascal–type growth. Double steps differ in that the terminal entry is duplicated, causing the final term to appear twice while preserving monotonicity.

Because the grammar forces expansive configurations to appear only within this minimal 5-block, extremal asymptotic growth is realized by repeated alignment of this block, making it sufficient to analyze this configuration. Thus for extremal 5-cycle analysis, the local maximal sequence (dominating per-pattern growth) is

$$\{s, d, s, d, d\}, \{s, d, s, d, d\},$$

enumerating occurrences of each admissible local pattern within a 5-cycle.

Admissible Local Patterns

Within A022921, only four local configurations occur:

$$(s, d), \quad (d, d), \quad (s, d, s), \quad (d, d, s).$$

The associated coefficient counts within a 5-cycle are:

Pattern	Associated coefficient count
s, d	2
d, d	1
s, d, s	1
d, d, s	1

D.2 Illustrative Extremal Computations

The generating rows evolve by a horizontal–sum rule (see Equation D.1) so that sequences of single steps exhibit ordinary Pascal–type growth. Double steps differ only in that the terminal entry of the row is duplicated, causing the final term to appear twice in the generating tuple while preserving monotonicity of the preceding entries.

These examples are chosen so that each step saturates the inequalities of Lemma 12, thereby realizing extremal growth. Equalities shown below should be interpreted as extremal upper-bound realizations rather than identities valid for all rows.

The following row-by-row computations realize extremal growth for the admissible local patterns and together determine the sharpest uniform bounds used in Lemma 23. Each example illustrates how a local pattern can saturate its maximal growth factor under the horizontal-sum dynamics. Let S denote the total sum of entries in the generating row prior to a given local pattern.

Example 1.

$$\textit{Single } a(10) = \underbrace{173}_{\text{sum:}=S} : \underbrace{1, 7, 25, 55, 85}_{S}$$

$$\textit{Double } a(11) = \underbrace{476}_{\text{sum}=3S} : \underbrace{1, 8, 33, 88}_{S} \underbrace{173}_{S}, \underbrace{173}_{S}$$

$$\textit{Single } a(12) = \underbrace{961}_{\text{sum}=9S} : \underbrace{1, 9, 42, 130}_{2S} \underbrace{303}_{3S}, \underbrace{476}_{4S}$$

$$\textit{Double } a(13) = \underbrace{2652}_{\text{sum}=33S} : \underbrace{1, 10, 52, 182}_{4S}, \underbrace{485}_{7S}, \underbrace{961}_{11S}, \underbrace{961}_{11S}$$

$$\textit{Double } a(14) = \underbrace{8045}_{\text{sum}=123S} : \underbrace{1, 11, 63, 245}_{8S}, \underbrace{730}_{15S}, \underbrace{1691}_{26S}, \underbrace{2652}_{37S}, \underbrace{2652}_{37S}$$

$$\textit{Single } a(15) = \underbrace{17637}_{\text{sum}=329S} : \underbrace{1, 12, 75, 320}_{16S}, \underbrace{1050}_{31S}, \underbrace{2741}_{57S}, \underbrace{5393}_{94S}, \underbrace{8045}_{131S}$$

The resulting maximal local ratios are

Pattern	Max Ratio
s, d	$3/1$
d, d	$123/33$
s, d, s	$9/3$
d, d, s	$329/123$

Example 2.

$$Single\ a(12) = \underbrace{961}_{\text{sum:=}S} : \underbrace{1, 9, 42, 130, 303, 476}_{S}$$

$$Double\ a(13) = \underbrace{2652}_{\text{sum=}3S} : \underbrace{1, 10, 52, 182, 485}_{S}, \underbrace{961}_{S}, \underbrace{961}_{S}$$

$$Double\ a(14) = \underbrace{8045}_{\text{sum=}13S} : \underbrace{1, 11, 63, 245, 730}_{2S}, \underbrace{1691}_{3S}, \underbrace{2652}_{4S}, \underbrace{2652}_{4S}$$

$$Single\ a(15) = \underbrace{17637}_{\text{sum=}37S} : \underbrace{1, 12, 75, 320, 1050}_{4S}, \underbrace{2741}_{7S}, \underbrace{5393}_{11S}, \underbrace{8045}_{15S}$$

$$Double\ a(16) = \underbrace{51033}_{\text{sum=}131S} : \underbrace{1, 13, 88, 408, 1458}_{8S}, \underbrace{4199}_{15S}, \underbrace{9592}_{26S}, \underbrace{17637}_{41S}, \underbrace{17637}_{41S}$$

$$Single\ a(17) = \underbrace{108950}_{\text{sum=}341S} : \underbrace{1, 14, 102, 510, 1968}_{16S}, \underbrace{6167}_{31S}, \underbrace{15759}_{57S}, \underbrace{33396}_{98S}, \underbrace{51033}_{139S}$$

The corresponding maximal local ratios are

Pattern	Max Ratio
s, d	3/1
d, d	13/3
s, d, s	341/131
d, d, s	37/13

Improved Extremal Alignment

Because the total multiplicative growth over a cycle is the product of the local growth factors, the geometric mean over one full cycle is maximized by selecting the maximal admissible factor at each local pattern occurrence. Consequently, once a uniform upper bound has been established for each admissible local pattern, the extremal global growth rate over a complete cycle is obtained by multiplying these maximal local factors according to their frequencies and taking the corresponding geometric mean.

Equivalently, because total growth over any admissible word is multiplicative in its local factors, the asymptotic per-symbol growth rate is the geometric mean of these *factors; hence the maximal rate over the entire language generated by the grammar is achieved by the word that maximizes this mean subject to the grammar constraints.*

Each example yields a valid (not necessarily sharp) upper bound on the maximal growth factor associated with each admissible local pattern. Since these bounds arise from the same uniform recurrence and apply globally to all admissible configurations, the true maximal growth factor for a given pattern is bounded above by the smallest recorded bound among all such valid upper bounds.

Because the recurrence is monotone with respect to the placement of delayed doubling, no other admissible configuration can exceed these extremal realizations. Accordingly, combining the sharpest bounds obtained across different initiating rows yields the tightest global envelope on admissible local growth.

Pattern	Max Ratio	Associated coefficient count
s, d	3	2
d, d	123/33	1
s, d, s	341/131	1
d, d, s	329/123	1

Consequently, the maximal geometric growth rate of a 5–cycle satisfies

$$\Big(3^2 \cdot (123/33) \cdot (341/131) \cdot (329/123)\Big)^{1/5} \approx 2.976 < 3. \tag{D.2}$$

This bound directly feeds into Theorem 16 and ensures that the per-symbol growth of any admissible infinite sequence under A186009 is strictly dominated by 3, as explained in the main text.

Conclusion. Every admissible growth pattern of A186009 is constrained by the grammar of A022921. Infinite concatenations of 5-cycles, though forbidden, saturate local growth constraints and provide a strict upper envelope. Equation (D.2) confirms that even this extremal envelope has geometric growth rate < 3, ensuring subdominance relative to the denominator $2^{B(i)}$.

E Horizontal-Sum Table for A186009

For reference, a portion of A186009 generated by the horizontal-sum construction is given below as per an algorithm derived from Winkler [9]. This table is included for illustration of the horizontal-sum mechanism; none of the density or growth bounds in the main text depend on these specific numeric values.

Table 1: A186009 Horizontal Sum Formulation

n	$a_1(n)$	$a_2(n)$	$a_3(n)$	$a_4(n)$	$a_5(n)$	$a_6(n)$	$a_7(n)$	$a_8(n)$	$a_9(n)$	$\sum a_i(n)$
1	1									1
2	1									1
3	1									1
4	1	1								2
5	1	2								3
6	1	3	3							7
7	1	4	7							12
8	1	5	12	12						30
9	1	6	18	30	30					85
10	1	7	25	55	85					173
11	1	8	33	88	173	173				476
12	1	9	42	130	303	476				961
13	1	10	52	182	485	961	961			2652
14	1	11	63	245	730	1691	2652	2652		8045
15	1	12	75	320	1050	2741	5393	8045		17637
16	1	13	88	408	1458	4199	9592	17637	17637	51033
17	1	14	102	510	1968	6167	15759	33396	51033	108950

There is a sequence closely related to A186009 called A100982. A186009 prepends the number 1 to the A100982 sequence. Along with the listing of initial terms for A100982 comes an algorithm which explains how to generate the entire sequence starting with an initial value of 1. As a result of its close similarity, this method can also be leveraged to generate all of the terms in A186009.

Theorem 2 [9] referenced in A100982 asserts that the elements in the sequence can be derived in a Pascal–triangle–type manner, beginning with an initial condition:

$$a(1) = a_1(1) = 1$$

The algorithm for $A100982(n)$ states that the k^{th} term of the n^{th} series component can be computed as follows:

$$a_{k+1}(n) = a_k(n) + a_k(n-1); k \in \mathbb{N}_0, n \in \mathbb{N}$$

The algorithm also stipulates that the last term in the sequence is repeated whenever $A022921(n-2) = 2$ (which can be derived from A020914 by taking the first differences between its terms). In doing so, this imposes the implicit constraint that this rule applies only when $n \geq 2$. Keeping in mind that repeated terms in A022921 occur only once $n \geq 2$, the recurrence is extended to $n = 1$ by treating the duplication condition as inactive at the boundary, producing an augmented vertical-sum expansion.

In rows without terminal duplication, this recurrence is exactly Pascal's binomial rule; duplication acts only at the boundary and therefore preserves the monotone, binomial-type growth of interior entries. This boundary-only modification is precisely why the extremal growth analysis in Appendix D reduces to *local pattern frequencies* rather than global row shape.

We reformulate this construction such that the first non-zero term in each row is always at index 1 and thus the formula henceforth (producing the same expansion components up to a reindexing of coordinates) is modified as follows.

Preserving $a(1) = a_1(1) = a_1(n) = 1$ for all $n \in \mathbb{N}$, the i^{th} term of the n^{th} A100982 element is:

$$a_i(n) = a_i(n-1) + a_{i-1}(n); i, n \in \mathbb{N}$$

This reindexing does not change any row sums or terminal-duplication positions; it only relocates coordinates. Using this formulation the initial term for all values of $n \in \mathbb{N}$ is always 1 rather than shifting out for each value of n. The other significant difference is that in this formulation the values for $a_i(n)$ appear horizontally as opposed to vertically. The doubling rule for the terminal entry remains unchanged in substance – namely that it occurs when $A022921(n-2) = 2$ - which is precisely the single/double grammar governing admissible growth in the preceding appendices, and hence the local multiplicative structure used in the extremal bounds.

Because A186009 is obtained from A100982 by prepending an initial row, all row indices shift by one. Consequently the terminal-duplication condition expressed in the original formulation as $A022921(n-2) = 2$ now becomes $A022921(n-3) = 2$ in the present indexing.

F Enumeration of Admissible Division Words

Purpose and Scope

This appendix provides an explicit, constructive procedure for enumerating admissible division words of fixed length under the prefix constraints governed by A020914. While none of the density or growth bounds in the main text depend on these constructions, the algorithm clarifies the combinatorial structure underlying A186009 and makes explicit the finite grammar inducing admissible division words implicit in the horizontal-sum formulation.

Prefix-Sum Formulation

Let a division word of length $m \geq 1$ be a tuple

$$(d_0, d_1, \ldots, d_{m-1}), \qquad d_i \in \mathbb{N},$$

where d_i records the number of divisions by 2 following the ith odd step. Define the prefix sums

$$H(i) = \sum_{j=0}^{i} d_j.$$

Admissible division words of length m are exactly those satisfying

$$H(i) < A020914(i+1) \quad (0 \leq i < m-1), \qquad H(m-1) = A020914(m).$$

The strict inequalities for $i < m-1$ suppress shorter stopping-time words, while equality in the final column ensures the total division count matches the minimal exponent associated with length m.

Thus admissible division words are determined by a bounded prefix-sum search with backtracking.

Constructive Enumeration Procedure

Fix a word length $m \geq 1$ and define m columns numbered $i = 0, \ldots, m-1$. Below each i calculate the division word grammar bounds $A(i+1) = A020914(i+1)$.

Then, under each column, place the exponent entries d_i of the division word. The left-to-right horizontal sum $H(i)$ is the sum of entries up to column i, where $H(i)$ must satisfy the admissibility bounds described above.

Iterative Search

Generation of the set of all division words of length m proceeds by iterating through the following steps until all admissible words have been found.

Set $i = 0$ and $H(-1) = 0$ as the initial boundary condition. Begin at step 1.

1. Set successive row entries $d_i = 1$ up to column $m-2$:
   (a) If $i < m-1$, set $d_i = 1$. Increment i and repeat Step 1.
   (b) Otherwise, proceed to Step 2.
2. Calculate the final entry d_{m-1} so that the horizontal sum matches $A(m)$:
   $$d_{m-1} = A(m) - H(m-2),$$
   record the resulting division word $(d_0, \ldots, d_{m-1})$. Continue to step 3.
3. Increment the $(i-1)^{th}$ column if this is admissible by the grammar:
   (a) If $H(i-1) < A(i) - 1$, copy the previous row entries up to $i-2$.
   Increment d_{i-1} by 1, and return to Step 1.
   (b) Otherwise, proceed to Step 4.
4. Check the termination condition for the division word search:
   (a) If i is greater than 1, decrement i by one and return to Step 3.
   (b) Otherwise, all admissible division words of length m have been generated. Stop.

Example: Division Words of Length $m = 5$

For $m = 5$, the bounds are

$$A020914(1{:}5) = (2, 4, 5, 7, 8).$$

All admissible division words must satisfy

$$H(0) < 2,\ H(1) < 4,\ H(2) < 5,\ H(3) < 7,\ H(4) = 8.$$

The search procedure yields the seven admissible words shown in Table 2.

Table 2: Admissible division words of length $m = 5$.

i	0	1	2	3	4
$A(i+1)$	2	4	5	7	8
Word	d_0	d_1	d_2	d_3	d_4
1	1	1	1	1	4
2	1	1	1	2	3
3	1	1	1	3	2
4	1	1	2	1	3
5	1	1	2	2	2
6	1	2	1	1	3
7	1	2	1	2	2

Thus $A186009(m+1) = 7$ counts admissible division words of length $m = 5$. The final word $(1, 2, 1, 2, 2)$ is extremal in the sense that its prefix sums approach the bounds $A020914(i + 1)$ as closely as possible for $i < m - 1$ while remaining admissible, and achieve equality at $i = m - 1$.

G A Binomial Lower Comparison for A186009

The sequence A047749 admits a horizontal-sum construction analogous to that of A186009, but with a stricter terminal rule: its doubling mechanism is governed by A000034$(n-2)$ rather than A022921. As a result, A047749 forbids *consecutive* doublings, while such events occur periodically in A186009. This structural restriction yields a natural comparison sequence.

Pointwise comparison

Because A047749 evolves under the same Pascal-type recursion but with a subset of the admissible doubling events, its row sums form a pointwise lower bound (up to index shift):

$$A047749(n) \leq A186009(n+1), \qquad n \geq 0.$$

The two sequences agree initially, but diverge once the first consecutive doubling occurs in A186009; thereafter the inequality is strict.

Closed form for A047749

Unlike A186009, the restricted doubling grammar of A047749 permits an exact description. Its terms split naturally into even and odd indices:

Even indices ($n = 2m$).

$$a(n) = \frac{1}{2m+1}\binom{3m}{m}.$$

Odd indices ($n = 2m+1$).

$$a(n) = \frac{1}{2m+1}\binom{3m+1}{m+1}.$$

Equivalently, using Gamma functions,

$$a(2m) = \frac{\Gamma(3m+1)}{\Gamma(2m+2)\Gamma(m+1)}, \qquad a(2m+1) = \frac{\Gamma(3m+2)}{\Gamma(2m+2)\Gamma(m+2)}.$$

Asymptotic growth

Let $\psi(x) = \Gamma'(x)/\Gamma(x)$ be the digamma function. Differentiating the logarithm of the Gamma representations gives

$$\frac{d}{dm} \ln a(2m) = 3\psi(3m+1) - 2\psi(2m+2) - \psi(m+1),$$
$$\frac{d}{dm} \ln a(2m+1) = 3\psi(3m+2) - 2\psi(2m+2) - \psi(m+2).$$

Using $\psi(x) = \ln x + O(1/x)$ as $x \to \infty$, both expressions have the same limit:

$$\lim_{m\to\infty} \frac{d}{dm} \ln a(n) = \ln\left(\frac{3^3}{2^2}\right) = \ln\left(\frac{27}{4}\right).$$

Thus

$$\lim_{n\to\infty} \frac{a(n+2)}{a(n)} = \frac{27}{4}, \qquad a(n) \sim C\left(\sqrt{\frac{27}{4}}\right)^n.$$

Implication for A186009

Since A047749 is generated by a subset of the admissible doubling events for A186009, it provides a constructive comparison sequence with explicit exponential growth rate

$$\sqrt{\frac{27}{4}} = \frac{3\sqrt{3}}{2} \approx 2.598.$$

Hence A186009 admits an explicit exponentially growing comparison sequence, and grows strictly faster once consecutive doubling regimes become active.

References

[1] Lothar Collatz. On the iteration of numerical functions. *Mathematische Zeitschrift*, 1937. In German.

[2] Jeffrey C. Lagarias. *The Ultimate Challenge: The 3x+1 Problem*, volume 55 of *Student Mathematical Library*. American Mathematical Society, 2010.

[3] Richard Terras. A stopping time problem on the positive integers. *Acta Arithmetica*, 30:241–252, 1976.

[4] Tomás Oliveira e Silva. Maximum excursion and stopping time record-holders for the 3x+1 problem. *Mathematics of Computation*, 68(225):371–384, 1999.

[5] Terence Tao. Almost all collatz orbits attain almost bounded values. *Forum of Mathematics, Pi*, 8:e14, 2020. doi: 10.1017/fmp.2020.7.

[6] C. J. Everett. Iteration of the number-theoretic function $f(2n) = n$, $f(2n+1) = 3n + 2$. *Advances in Mathematics*, 25:42–45, 1977.

[7] Günther J. Wirsching. *The Dynamical System Generated by the* $3n+1$ *Function*, volume 1681 of *Lecture Notes in Mathematics*. Springer, 1998.

[8] Marc Chamberland. An update on the 3x+1 problem. *Butlletí de la Societat Catalana de Matemàtiques*, 18(1):19–45, 2003. URL `https://chamberland.math.grinnell.edu/papers/3x_survey_eng.pdf`. Survey on symbolic encodings and structural approaches to the $3x$+1 problem.

[9] Mike Winkler. The algorithmic structure of the finite stopping time behavior of the $3x + 1$ function. *arXiv e-prints*, arXiv:1709.03385, November 2017. Version 3.